

$$M^1 = M \cap \mathrm{Sp}(V, \psi)$$

$$\mathrm{Sp}(V, \psi)$$

$$\dim 9 \subset \dim 36$$

$$N$$

$$\text{index } 2$$

$$G \cdot A_3$$

$$\text{index } 3$$

$$G = \mathrm{Hg}(X)$$

dim of the algebraic classes in $V^{\otimes 2m}$				Det, r_1 algebraic	exceptional classes algebraic
X^4 $m = 2$	X^6 $m = 3$	formula $c = 2, 5$	$\zeta \in M^1$		
3	15	$(2m - 1)(2m - 3) \cdots 1$	yes	no	no
4	35	$(c^3 + 3c^2 + 2c)/6$	yes	yes	no
4	45	$(c^3 + 2c)/3$	yes	yes	no
8	125	c^3	no	yes	yes

$M^1 = G \Leftrightarrow$ the Hodge conjecture for every $X^n \Leftrightarrow$ the two exceptional classes of $X \times X$ are algebraic
 $\Leftrightarrow$ some algebraic class on some power of X is not fixed by ζ

